THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 7: **Temperature and its Measurement** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A clinical thermometer shows 38.5 °C. The person has: 1

(a) normal temperature **(b)** low temperature
(c) fever **(d)** no temperature

2. Why does the reading of a clinical thermometer stay after it is removed from the mouth? 1

(a) because of the kink **(b)** because of the scale
(c) because of the glass **(d)** because of the bulb

3. Temperature of a cup of tea is 80 °C, the room is 30 °C. After a long time, the tea's temperature will be about: 1

(a) 80 °C **(b)** 100 °C **(c)** 50 °C **(d)** 30 °C

4. Mercury level before measuring should be: 1

(a) at 100 °C **(b)** at 42 °C **(c)** below 35 °C **(d)** at 40 °C

5. Why should a lab thermometer not be held by the bulb while reading? 1

(a) it will break **(b)** bulb is sensitive to body heat
(c) it is cold **(d)** it is slippery

6. Water at 20 °C and water at 60 °C are mixed. The mixture could be: 1

(a) 40 °C **(b)** 80 °C **(c)** 10 °C **(d)** 70 °C

7. A hot iron rod and a hot wooden rod, both at 60 °C, are touched. Which feels hotter? 1

(a) wood **(b)** iron **(c)** both equal **(d)** neither

8. Why does a person feel colder after bathing in air than in water? 1

(a) air is wet **(b)** water is cold
(c) clothes are tight **(d)** evaporation takes heat away

9. In summer, a metal door handle feels hotter than a wooden door because: 1

(a) metal is heavier
(b) the metal conducts heat to the hand faster
(c) wood is shiny
(d) wood is colder

10. A student measures water temperature as 95 °C and later 60 °C. The heat flows from water to: 1

(a) surroundings **(b)** nowhere
(c) the thermometer to the water **(d)** water from surroundings

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A clinical thermometer reading is 39 °C. Interpret it. 2

12. Why do we wear several layers instead of one thick layer in winter? 2

13. Describe two uses of the thermos flask. 2

14. Why should the bulb of a laboratory thermometer not touch the bottom of the beaker? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A student reads a thermometer as 70 °C for hot water and 25 °C for the room. Describe what will happen to the water temperature in 2 hours. 3

16. Why are the walls of a thermos flask silvered? 3

17. Why is a thin cotton cloth preferred in summer? 3

OR

Design an experiment to compare the conduction of copper and iron rods. 3

18. Explain why an iron nail sunk in hot water feels hot at once but a wooden stick feels warm slowly. 3

OR

Why does a hot iron plate burn the hand but a hot wooden plate at the same temperature may not? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A student heats water in a beaker and records the temperature every 2 minutes: 30, 38, 46, 54, 62 °C. Find the rise per 2 minutes and the likely temperature after 10 minutes. 5

OR

Explain the idea of thermal equilibrium with the example of mixing hot and cold water. 5

20. Explain how a clinical thermometer works, why it has a kink and why it should not be heated. 5

OR

Why is it unsafe to use a lab thermometer in the mouth? 5

END OF QUESTION PAPER — Total: 40 Marks

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